

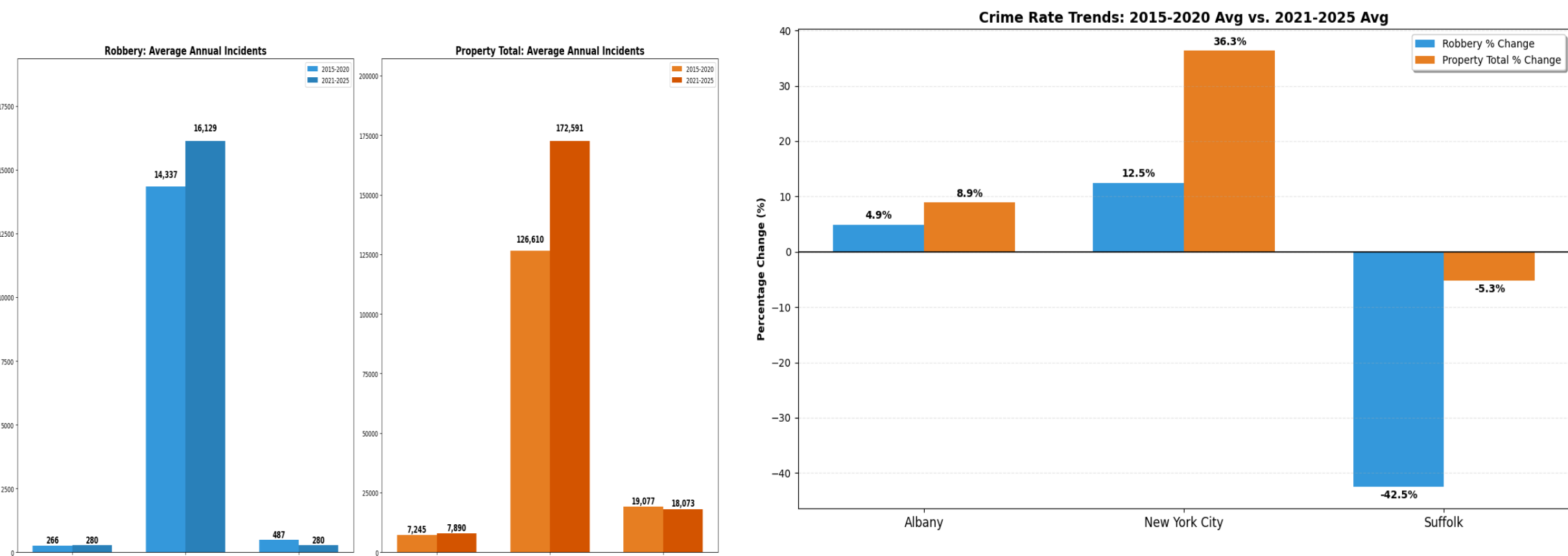


New York Crime Patterns

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Abstract

Our goal was to observe and visualize crime trends among the three highly populated counties in New York State: Albany, Suffolk and New York City (five boroughs combined into one). We wanted to see which county has the highest percentage of robbery and total property crime? Additionally, how did robbery and total property crime rates change in the five years following 2020 compared to the five years preceding 2020? We found that crime patterns appear to change due to societal issues at the time. Based on the data, we infer that the effects of the Covid 19 pandemic on the economy influenced the property and robbery crime fluctuations. In general, recent crime is on a decline in comparison to the 1990s.



Original Graphs we used to evaluate our research questions

Methods [or Process]

- Our group wanted to tackle a relevant societal issue: Did Covid 19 increase crime rates?
- With this idea we planned to use New York’s major cities with plentiful data
- We first sorted and cleaned our data to make a relevant graph which helped us determine if our problem had cause for a deeper analysis (Graphs seen above)
- After the original graphs we found that there could be a correlation, which allowed us to implement machine learning algorithms(KNN, Decision Tree, Linear Regression, and Random Forrest Regression) to visualize the trends(graphs in results)

Results

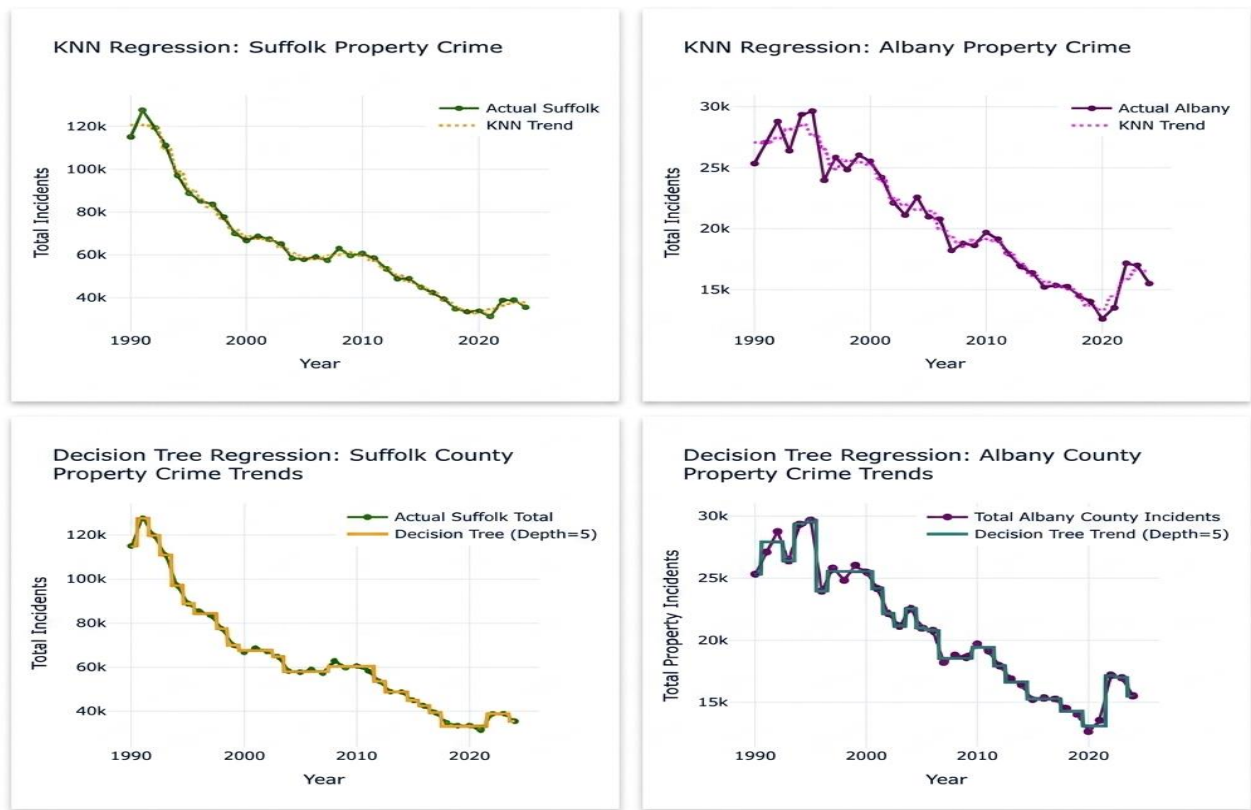
What we were able to learn:

From our graphs we were able to analyze a correlation between rising crime rates and the 5 years post 2020.

This correlation could lead us to infer that post covid economy could be a link although can’t fully be proven.

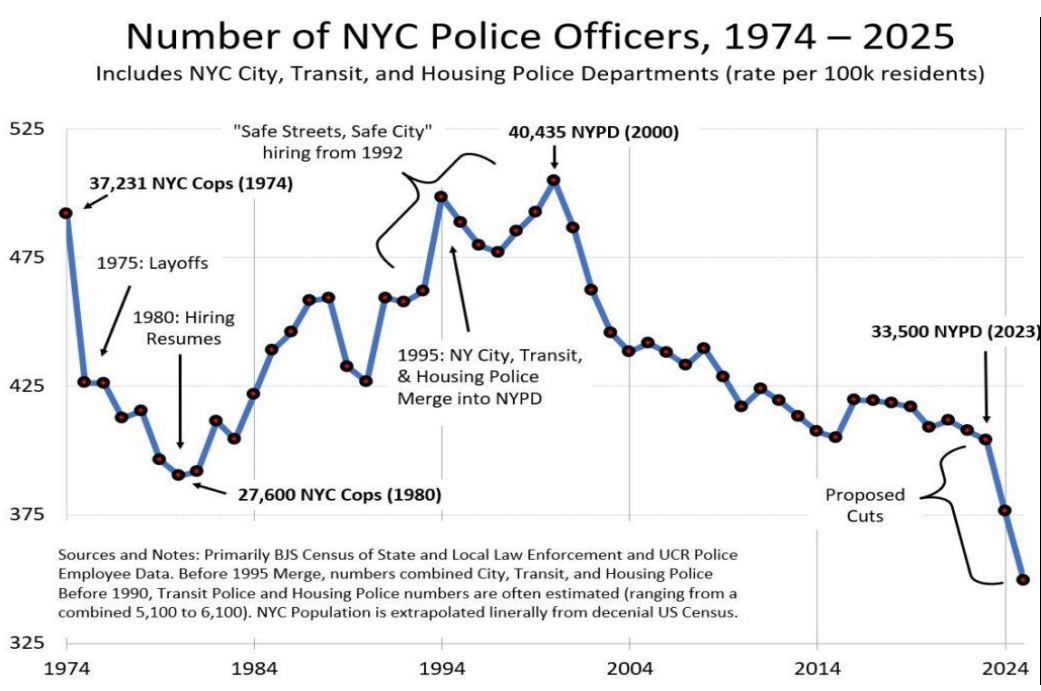
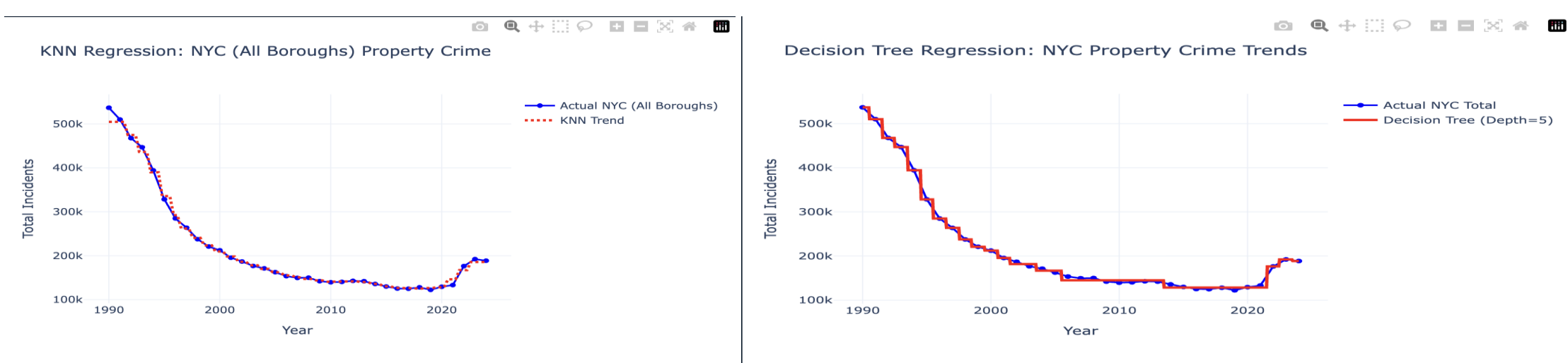
What we can visibly see is the spikes in crime rate both in Property and Robbery.

Although an outlier to our results is Suffolk, as they seem to not increase as compared to the counties.



Here we can see KNN and Decision Tree Algorithms on Both Albany and Suffolk Counties.

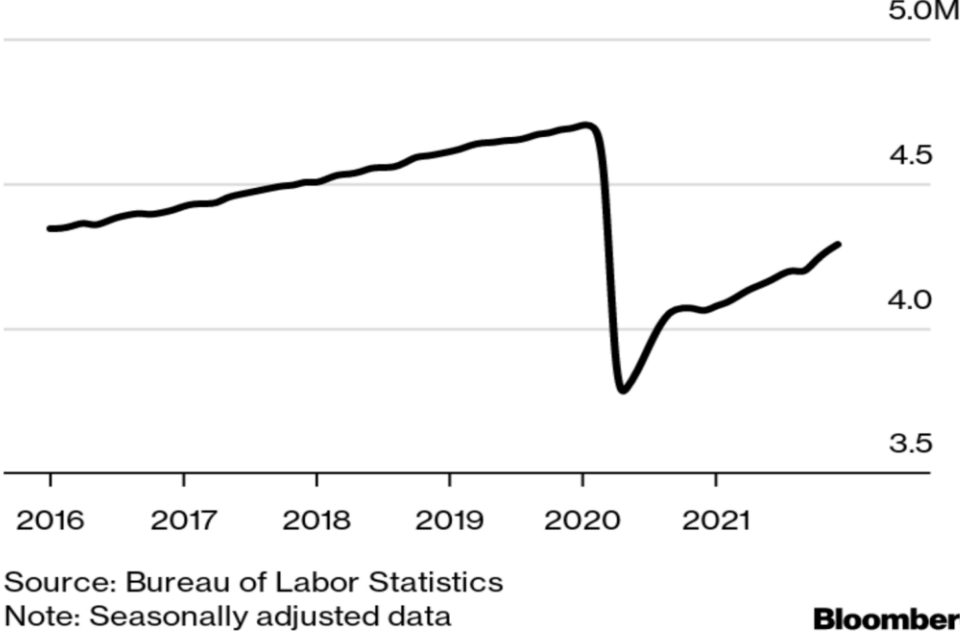
Significance



This graph shows the decline of police officers in New York. Which overlaps with the increase in crime. Our data and models could be a helpful presentation, as a counter argument to future proposed police cuts. Aswell as to visualize the cause and effect relationship of decreasing officers to crime rate.

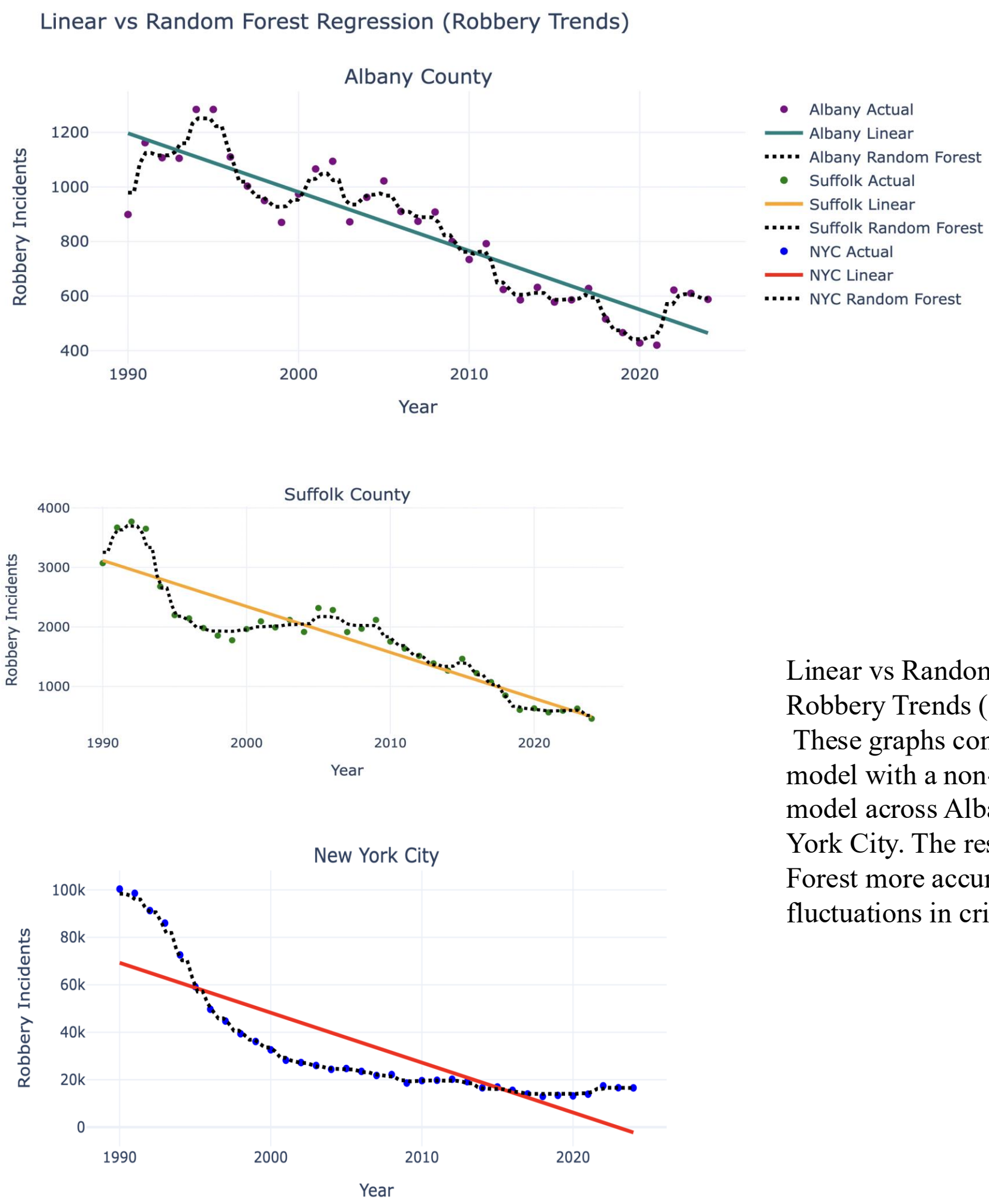
Pay Check Blues
New York City's employment recovery has been less than speedy

✓ Total nonfarm payrolls



Source: Bureau of Labor Statistics
Note: Seasonally adjusted data

Following 2020, robbery trends show slight increases or fluctuations compared to previous years. This indicates that recent societal or economic changes may have disrupted long-term downward trends in violent crime.



Linear vs Random Forest Regression of Robbery Trends (1990–2025)
These graphs compare a simple linear model with a non-linear Random Forest model across Albany, Suffolk, and New York City. The results show that Random Forest more accurately captures fluctuations in crime trends over time.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# 1. Load Data
file_path = '/Users/andrewsanft/Downloads/Index_Crimes_by_County_2'
data = pd.read_csv(file_path)
data.columns = data.columns.str.strip()

# 2. Identify Property Column
prop_col = None
for c in data.columns:
    if 'Property' in c and 'Total' in c:
        prop_col = c
        break

# 3. Filter for Aggregate Data
df = data[cols]
df['County'] = df['County'].str.replace(' ', '_').str.lower()
nyc_boroughs = ['Bronx', 'Kings', 'New York', 'Queens', 'Richmond']
target_counties = ['Albany', 'Suffolk']

# 4. Period Logic
def get_period(period):
    if period == '2015-2020':
        return '2015-2020'
    elif period == '2021-2025':
        return '2021-2025'
    else:
        return None

# 5. Force Numeric Cleaning (Crucial for Labels)
def clean_numeric(x):
    try:
        return float(x)
    except:
        return None

# 6. Visualization
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(20, 10))

# Albany County
ax1.set_xlabel('Year')
ax1.set_ylabel('Robbery Incidents')
ax1.plot(actual_data['Year'], actual_data['Property Total'], color='red', marker='x')
ax1.plot(predicted_data['Year'], predicted_data['Property Total'], color='blue', marker='o')
ax1.legend(['Actual', 'KNN'])

# Suffolk County
ax2.set_xlabel('Year')
ax2.set_ylabel('Robbery Incidents')
ax2.plot(actual_data['Year'], actual_data['Property Total'], color='red', marker='x')
ax2.plot(predicted_data['Year'], predicted_data['Property Total'], color='blue', marker='o')
ax2.legend(['Actual', 'KNN'])

# New York City
ax3.set_xlabel('Year')
ax3.set_ylabel('Robbery Incidents')
ax3.plot(actual_data['Year'], actual_data['Property Total'], color='red', marker='x')
ax3.plot(predicted_data['Year'], predicted_data['Property Total'], color='blue', marker='o')
ax3.legend(['Actual', 'KNN'])

# Final Layout Adjustments
fig.update_layout(
    title='Property Crime Trends Across NY Regions',
    template='plotly_white',
    showlegend=True,
    hovermode='x unified'
)

# Set common axes titles
fig.update_xaxes(title_text='Year')
fig.update_yaxes(title_text='Total Incidents')

fig.show()
```